43. $\Delta T_f = K_f . m$

ie $\Delta T_f \alpha m$

$$m_{CH_3-OH} = \frac{\left(\frac{10}{32}\right)}{0.1} = 3.125m$$

$$m_{C_2H_5-OH} = \frac{\left(\frac{20}{46}\right)}{0.2} = 2.174m$$

$\therefore$ depression in freezing point is more in methanol solution and it will have lower freezing point.

44. In 10^{-4}M K_2SO_4 solution, there are 10^{-4} moles of potassium sulphate.

K_2SO_4 molecule contains 3 ions ($2K^+$ and $1So_4^{2-}$)

1 mole of K_2SO_4 contains $3 \times 6.023 \times 10^{23}$ ions

10^{-4} mole of K_2SO_4 contains $3 \times 6.023 \times 10^{23} \times 10^{-4}$ ions

$= 18.069 \times 10^{19}$

45. $(k_H)_{bonzene} = 4.2 \times 10^{-5}$ mm Hg

Solubility of methane = ?

P = 750mm Hg P = 840 mm Hg

According to Henrys Law,

$P = K_H . x_{\text{in solution}}$.

750mm Hg = 4.2×10^{-5} mm Hg . $x_{\text{in solution}}$

$$\Rightarrow x_{\text{in solution}} = \frac{750}{4.2 \times 10^{-5}}$$

i.e, solubility = 178.5×10^5

similarly at P = 840 mm Hg

$$\text{solubility} = \frac{840}{4.2 \times 10^{-5}}$$

$$= 200 \times 10^{-5}$$

46. $\Delta T_f = 0.093^oC = 0.093K$

m = ?

$K_f = 1.86K$ Kg mol^{-1}

$\Delta T_f = K_f . m$

$$\therefore m = \frac{\Delta T_f}{K_f}$$

$$= \frac{0.093K}{1.86\ K\ Kgmol^{-1}}$$

$= 0.05$ mol Kg^{-1}

$= 0.05m$

47. $P^0_{C_6H_6} = 640$mm Hg

$W_2 = 2.2$ g (non volabile solute)

$W_1 = 40$ g (benzene)

$P_{solution} = 600$ mm Hg

M_2 = ?

$$\frac{P^o - P}{P^0} = x_2$$

$$\frac{640 - 600}{640} = \frac{n_2}{n_1 + n_2} \quad [\because n_1 >> n_2 ; n_1 + n_2 \approx n_1]$$

$$\frac{40}{640} = \frac{n_2}{n_1}$$

$$0.0625 = \frac{W_2 \times M_1}{M_2 \times W_1}$$

$$M_2 = \frac{2.2 \times 78}{0.0625 \times 40}$$

$= 68.64$ g mol^{-1}

UNIT -10

CHEMICAL BONDING

KEY FOR MCQs

1. Compound — No. of valance electron on the central atom

XeF_2	10
$AlCl_3$	6
SF_6	12
SCl_2	8

(option d)

2. $\overset{\cdot\cdot}{\underset{A}{O}} = C = \overset{\cdot\cdot}{\underset{B}{O}}$

Key answers -jagan.indd 303 22-01-2021 13:23:34